2025.10.29. (Distributions (Schwarz)) 广义函数

Ω domain in $\mathbb{R}^n$ → connected

$\mathcal{D}(\Omega) = C_c^\infty(\Omega) = \{u \in C^\infty(\Omega) \mid \operatorname{supp} u \text{ is cpt} \subseteq \Omega\}$.

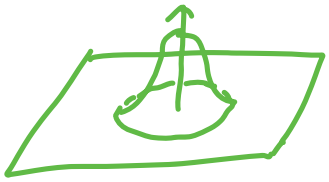
$\operatorname{supp} u = \overline{\{x \in \Omega \mid u(x) \neq 0\}}$, $u \in \mathcal{D}(\Omega)$ is compactly supported

e.g.1 The standard mollifier (光滑化子)

$$j(x) = \begin{cases} Ce^{\frac{1}{|x|^2-1}}, & x \in \mathbb{R}^n, |x|<1 \\ 0, & x \in \mathbb{R}^n, |x| \geq 1 \end{cases}$$

C is chosen s.t. $\int_{\mathbb{R}^n} j(x)dx = 1$

$\boxed{j(x) \in C_c^\infty(\mathbb{R}^n)}$



pf: Let $f(t) = \begin{cases} Ce^{\frac{1}{t-1}}, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$ $j(x) = f(|x|)$

It suffices to show $f \in C_c^\infty(\mathbb{R}^+)$.

$f \in C(\mathbb{R}^+)$, $f'(1+) = 0$, $f'(1-) = \lim_{t\to 1^-} \frac{f(t)-f(1)}{t-1} = \lim_{t\to 1^-} \frac{Ce^{\frac{1}{t-1}}}{t-1} = \lim_{s\to-\infty} Ce^s \cdot s = 0$

$\Rightarrow f'(1) = 0$

$f''(1+) = 0$, $f''(1-) = \lim_{t\to 1^-} \frac{f'(t)-f'(1)}{t-1} = \lim_{t\to 1^-} \frac{-Ce^{\frac{1}{t-1}}}{(t-1)^3} = \lim_{s\to-\infty} -Ce^s s^3 = 0$.

$\forall n \geq 1$, $f^{(n)}(1+) = 0$, $f^{(n)}(1-) = 0 \Rightarrow f^{(n)}(1) = 0$, $f \in C_c^\infty(\mathbb{R}^+)$

Q: $\forall \Omega \subseteq \mathbb{R}^n$ domain, open, Is $\mathcal{D}(\Omega) \neq \emptyset$?

Take $a \in \Omega$, $\varepsilon > 0$ small s.t. $\overline{B_\varepsilon(a)} \subseteq \Omega$, Let $j_{\varepsilon,a}(x) = \frac{1}{\varepsilon^n} j(\frac{x-a}{\varepsilon})$

$\operatorname{supp} j_{\varepsilon,a}(x) = \overline{B_\varepsilon(a)} \subseteq \Omega$. $\int_{\mathbb{R}^n} j_{\varepsilon,a}(x)dx = \int_{B_\varepsilon(a)} \frac{1}{\varepsilon^n} j(\frac{x-a}{\varepsilon})dx = \int_{B_1(0)} \frac{1}{\varepsilon^n} j(y)\varepsilon^n dy = 1$

Def! (Distribution)

A distribution on $\mathcal{D}(\Omega)$ is a linear functional $f: \mathcal{D}(\Omega) \to \mathbb{R}$.

s.t. f is continuous on $\mathcal{D}(\Omega)$ in the following sense: $\phi \mapsto \langle f, \phi \rangle$ or $f(\phi) \in \mathbb{R}$

$\forall \{\phi_k\}_{k=1}^\infty \subseteq \mathcal{D}(\Omega)$ with

(1) $\operatorname{supp} \phi_k \subseteq$ fixed compact $K \subseteq \Omega$.

(2) $\forall \alpha=(\alpha_1,\cdots,\alpha_n)\in\mathbb{Z}^n,\ \alpha_i\geq 0,\ |\alpha|=\sum_{k=1}^n \alpha_k,\ \lim_{k\to\infty}\|D^\alpha \phi_k\|_{L^\infty(\Omega)}=0.$

We have $\langle f,\phi_k\rangle\to 0\ (k\to\infty)$. f is continuous $\Leftrightarrow$ f is continuous with 0.

Def2 $\mathcal{D}'(\Omega)$ = set of all distribution on $\mathcal{D}(\Omega)$

e.g.2 $f\in L^1_{loc}(\Omega)$ $\forall \Omega'\subseteq\Omega$, $\overline{\Omega'}$ is compact & $\overline{\Omega'}\subseteq\Omega$. $\Rightarrow \int_{\overline{\Omega'}}|f|dx<\infty$

$\Omega'\subset\subset\Omega$.

$F\in\mathcal{D}'(\Omega)$ as: $\langle F,\phi\rangle=\int_\Omega f\cdot\phi dx\ \ \forall\phi\in\mathcal{D}(\Omega)$

Take $\{\phi_k\}_{k=1}^\infty\subseteq\mathcal{D}(\Omega)$ with ① and ②.

$|\langle F,\phi_k\rangle|=|\int_\Omega f\cdot\phi_k dx|\leq \|\phi_k\|_{L^\infty(\Omega)}\int_{\text{supp}\phi_k} f\,dx\to 0\ (k\to\infty)$.

$\Rightarrow \langle F,\phi_k\rangle\to 0\ (k\to\infty)$. F is induced by f ($F=f$).

e.g.3 Suppose $f\in\mathcal{D}'(\Omega)$, $g\in C^\infty(\Omega)$. Define $\langle fg,\phi\rangle=\langle f,g\phi\rangle$

$\forall\phi\in C_c^\infty(\Omega)\Rightarrow f\cdot g,\ g\cdot f$ are distribution.

(distribution $\cdot\ C^\infty$ = distribution)

e.g.4 Fix $a\in\Omega$. define $\delta_a:\mathcal{D}(\Omega)\to\mathbb{R}$, $\langle\delta_a,\phi\rangle=\phi(a)$ $\delta_a\in\mathcal{D}'(\Omega)$

Take $\{\phi_k\}_{k=1}^\infty$ with ① and ②. $|\langle\delta_a,\phi_k\rangle|=|\phi_k(a)|\leq\|\phi_k(a)\|_{L^\infty(\Omega)}\to 0$.

δ_a is the Dirac measure on $\mathbb{R}^n$ giving unit mass to the point a.

Def3 (Convergence)

Let $\{f_k\}_{k=1}^\infty\subseteq\mathcal{D}'(\Omega)$, $f\in\mathcal{D}'(\Omega)$, we say $f_k\to f\ (k\to\infty)$.

if $\forall\phi\in\mathcal{D}(\Omega)$, $\langle f_k,\phi\rangle\to\langle f,\phi\rangle\ (k\to\infty)$

Thm1 (Spiky Thm)

Let $\{f_k\}\subset L^1(\Omega)$ with

(1) $\{f_k\}$ concentrates at $a\in\Omega$ as $k\to\infty$: $\forall\delta>0$, $\int_{\Omega\setminus B_\delta(a)}|f_k|dx\to 0\ (k\to\infty)$.

(2) $\|f_k\|_{L^1(\Omega)}\leq M\ \ \forall k\geq 1$, $M>0$

(3) $\int_\Omega f_k\,dx \to A \quad (k\to\infty)$

Then f_k (understood as distr's) $\longrightarrow A\delta_a \quad (k\to\infty)$

Pf: Just to show $\forall \phi\in \mathcal{D}(\Omega)$, $\langle f_k,\phi\rangle \to \langle A\delta_a,\phi\rangle$

$$\begin{array}{ccc} \langle f_k,\phi\rangle & \to & \langle A\delta_a,\phi\rangle \\ \| & & \| \\ \int_\Omega f_k\phi\,dx & \to & A\phi(a) \\ \| & & \\ \int_\Omega f_k(\phi(x)-\phi(a))\,dx + \int_\Omega f_k\phi(a)\,dx & \to & A\phi(a). \end{array}$$

$\int_\Omega f_k(\phi(x)-\phi(a))\,dx \to 0$

$\forall \varepsilon>0, \exists \delta>0$, s.t. $|x-a|<\delta \Rightarrow |\phi(x)-\phi(a)|<\varepsilon$

$$\left|\int_\Omega f_k\cdot(\phi(x)-\phi(a))\,dx\right| = \left|\int_{B_\delta(a)} f_k\cdot(\phi(x)-\phi(a))\,dx + \int_{\Omega\setminus B_\delta(a)} f_k(\phi(x)-\phi(a))\,dx\right|$$

$$\le M\varepsilon + 2\|\phi(x)\|_{L^\infty(\Omega)}\cdot\int_{\Omega\setminus B_\delta(a)}|f_k|\,dx$$

$\overline{\lim_{k\to\infty}}\left|\int_\Omega f_k(\phi(x)-\phi(a))\,dx\right| \le M\varepsilon$, $\varepsilon\to 0^+ \Rightarrow \int_\Omega f_k(\phi(x)-\phi(a))\,dx \to 0 \quad (k\to\infty)$

e.g.5 Fix $a\in\Omega$. then $j_{\varepsilon,a}(x)\to\delta_a \quad (\varepsilon\to 0)$ in the sense of distr's.

By Spiky Thm, $\int_\Omega j_{\varepsilon,a}(x)\,dx = 1$, $M=1$, $A=1$. $\forall\delta>0$. $B_\delta(a)\subseteq\Omega$.

$\int_{\Omega\setminus B_\delta(a)}|f_k|\,dx = 0$ for $(\varepsilon<\delta) \Rightarrow j_{\varepsilon,a}(x)\to\delta_a \quad (\varepsilon\to 0)$.